

4. Classification

Chris Reudenbach

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Tasks

1. **Set AOI**
Load Burgwald AOI, transform to EPSG:4326, merge to one polygon.
2. **Use project directory structure**
Sentinel-2 data only in:
data/raw/AOI_Burgwald/s2/
Training areas in:
data/raw/AOI_Burgwald/train_areas_burgwald.gpkg
3. **Create training areas**
Digitise polygons inside the AOI, assign attribute `class`.
4. **Run STAC query**
Use `intersects = AOI`,
`datetime = 2018-05-01 to 2022-12-31`,
send as **POST request**, then call `items_fetch()`.
5. **Select suitable dates**
Compute per-date scene count + mean cloud cover,
rank by cloud cover, choose two low-cloud dates (2018 and 2022).
6. **Build predictor stacks**
For both dates: request raw bands + kNDVI + SAVI + EVI,
merge into a raster stack, save as:
data/processed/pred_stack_2018.tif
data/processed/pred_stack_2022.tif

7. Check consistency

Ensure identical extent, resolution, and layer order in both stacks.

8. Document step